

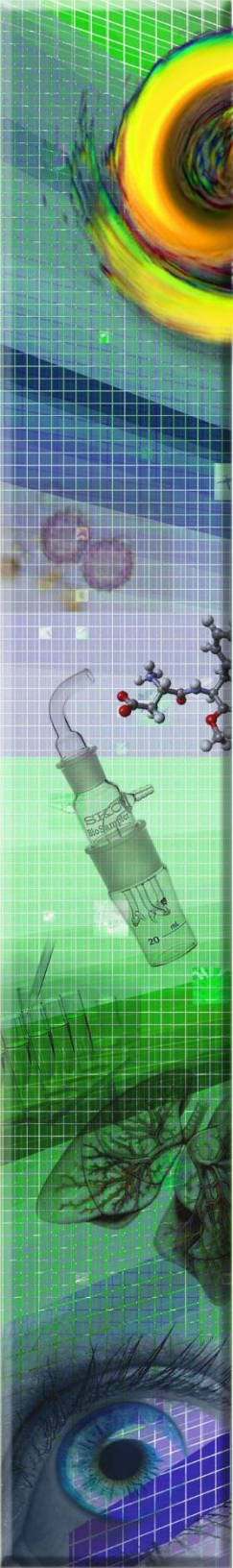
ARE Labs Inc.

Title: Foaming Cleanser Spray Coverage Testing

Client: Bragg & Bevier LLC

Project #: 11048.10

1/8/2026



Testing Overview

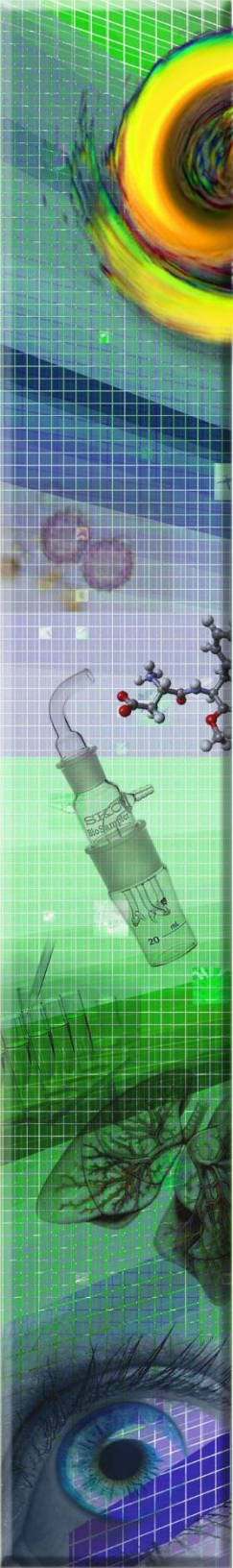
- Purpose of testing was to determine the spray coverage of the product inside a 96-gallon trash bin.
- 4 height levels in the trash bin were selected, including the interior lid surface
- Mounted 16 1"x1" paper filter coupons (4 per height including 4 on interior lid surface) to 96-gallon bin interior walls using adhesive.
- Initial mass of the spray can, and test coupons were recorded.
- Placed aerosol can upright at the center of the bin floor using double sided tape to secure it.
- Installed internal lighting and video recording device.
- Actuated can for full discharge with the lid closed.

$$\text{Surface Deposition (mg/in}^2\text{)} = \frac{\Delta\text{Mass}}{\text{Area}}$$

Testing Overview

- Allowed foam to settle for 20 seconds.
- Post-testing mass of the coupons was taken, and the surface deposition was calculated.
- Removed coupons and placed them in labeled containers.
- Post-testing mass of the spray can was recorded.





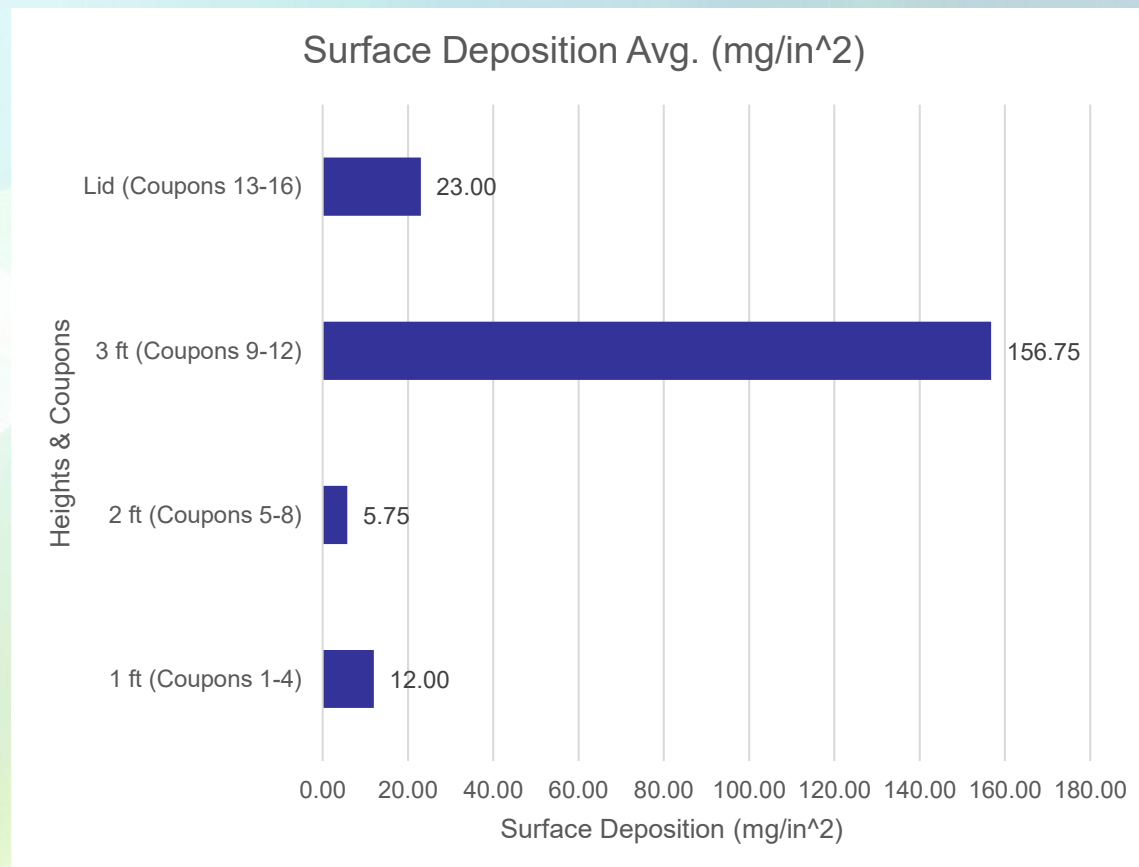
Foaming Cleanser Spray Coverage Results

Results

4 coupons at each height.

Initial Mass of Spray (no lid)	Post-Testing Mass
156.82g	65.18g

# ID	Initial Mass Avg. (mg)	Post-testing Mass Avg. (mg)	Surface Deposition Avg. (mg/in ²)
1 ft (Coupons 1-4)	64.25	76.25	12.00
2 ft (Coupons 5-8)	65.00	70.75	5.75
3 ft (Coupons 9-12)	63.00	219.75	156.75
Lid (Coupons 13-16)	71.50	94.50	23.00



Mid-Point

Coupons on the 2ft level.

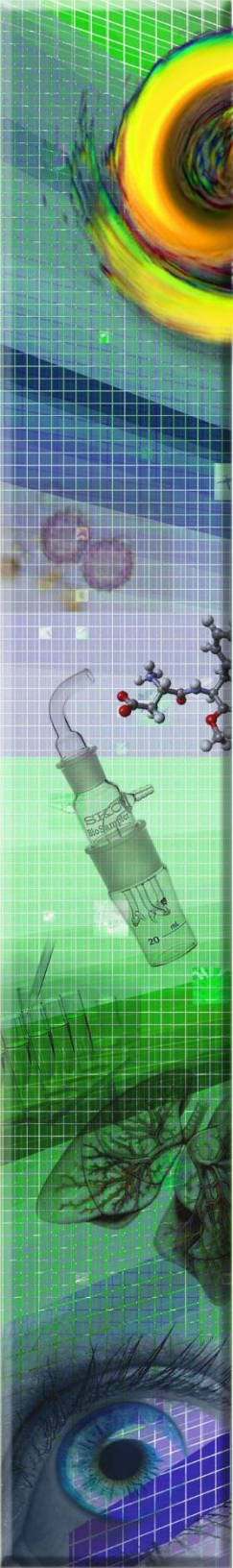
Coupons on the 1ft level.

Coupons at the 3ft level are out of frame.

End-Point



Since the spray was fan shaped it mainly only covered one side and hardly anything was got outside what is displayed in the circle.

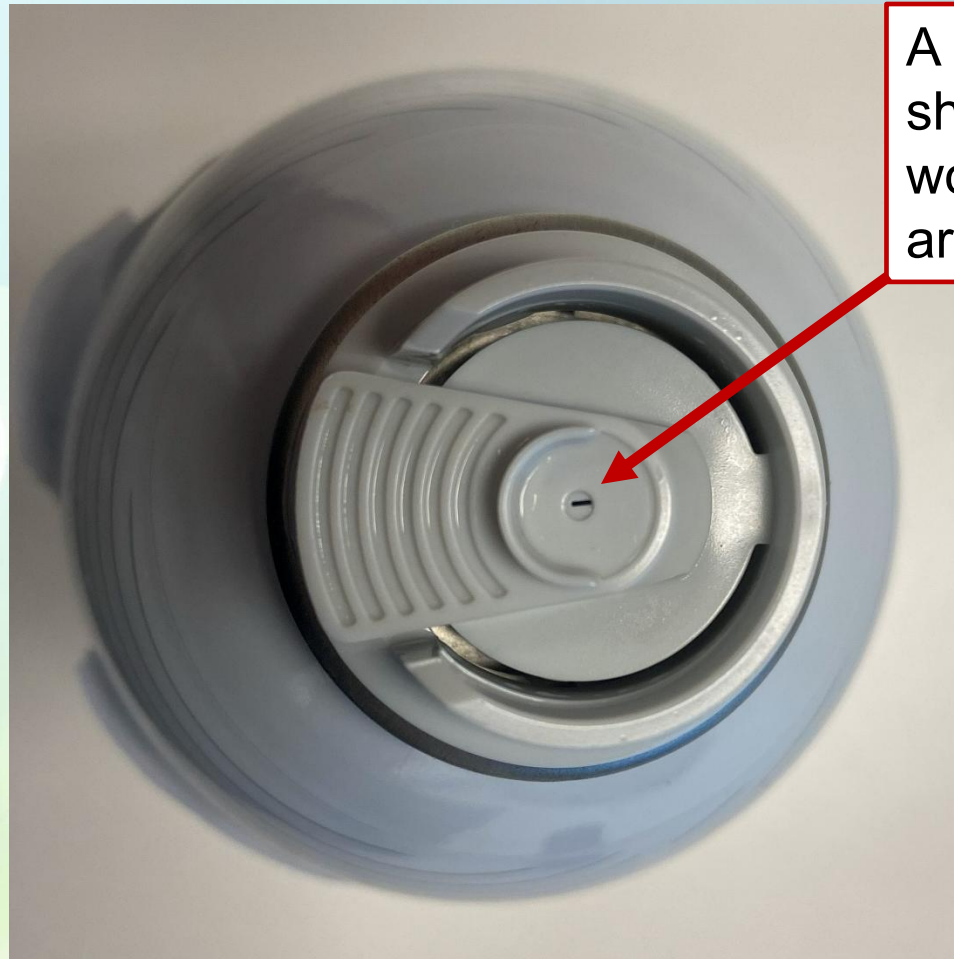


Summary

- The 2nd spray can had more output than the can used in the 1st attempt.
- The can sprayed high enough to reach the interior lid surface.
- The contents inside the can had a nice peppermint smell.
- The fan shaped spray covered half the trash can.
- Since the can sprays in a fan shape 5 of the 16 coupons were the only ones to get wet. 1 at the 1ft level, 1 at the 2ft level, 2 at the 3ft level, and 1 at the lid.

Discussion

- We noticed the shape of the nozzle being a line.
- A round nozzle that shoots in a cone shape rather than a line that shoots in fan shape would cover more area.



A round nozzle that shoots in a cone shape would improve the area the spray covers.